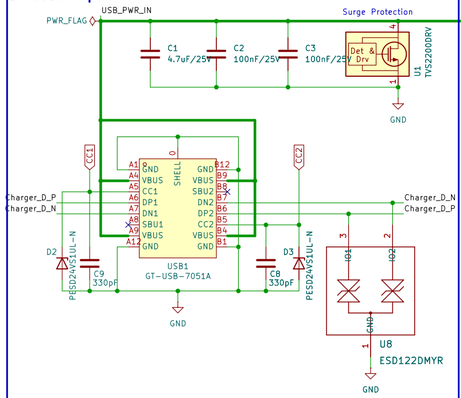
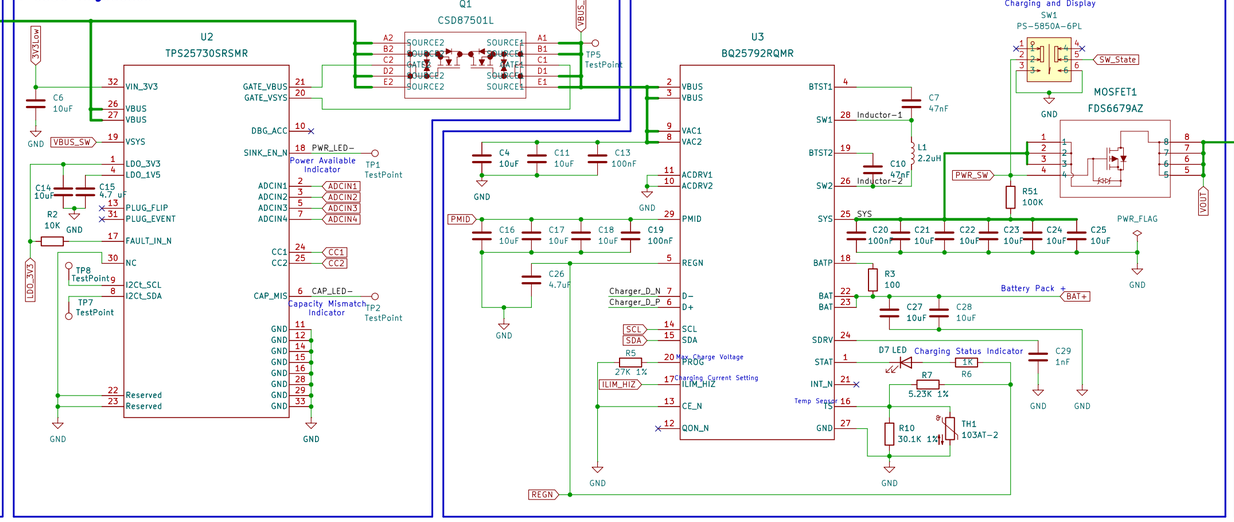
1. USBC Input

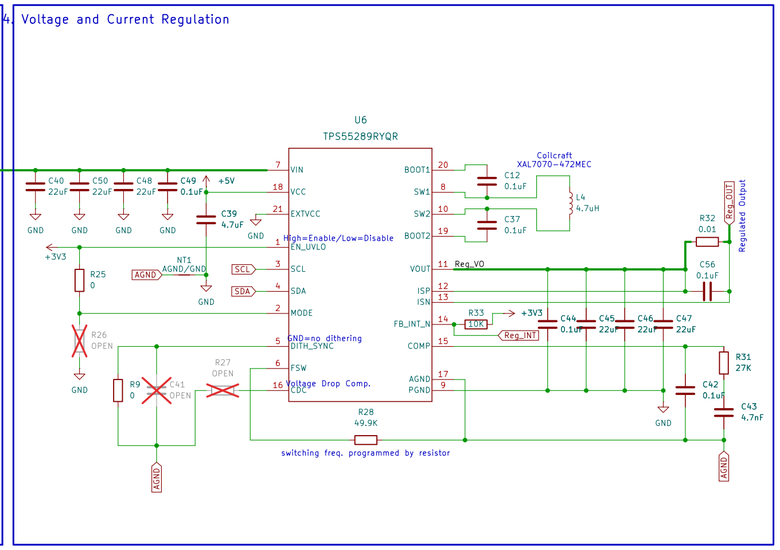


2. USBC Negotiation

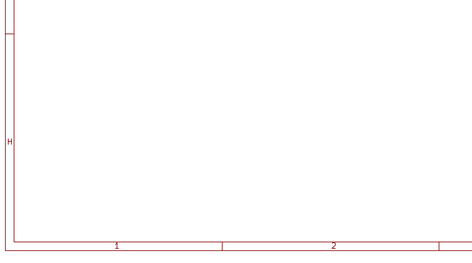


3. Battery Charging and Output

4. Voltage and Current Regulation



3.3V Regulator

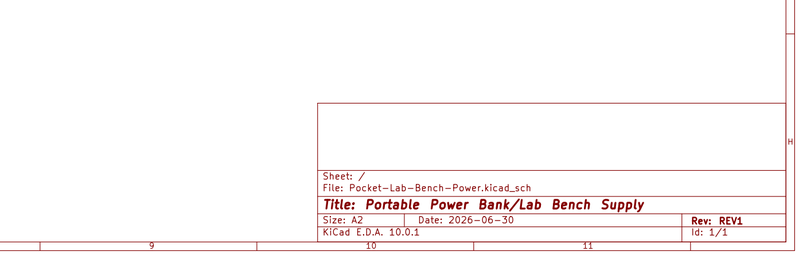


TPS25730 Configuration

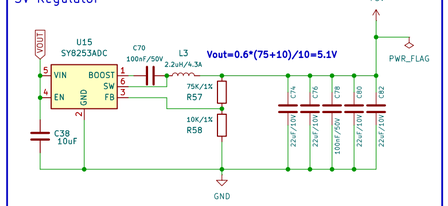
BQ25792 Configuration

Battery Gauge

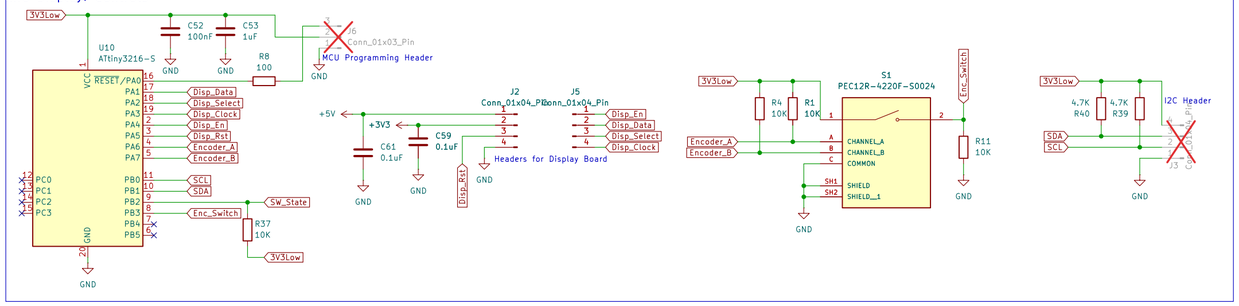
5. Output Gauge, Surge/Polarity protection



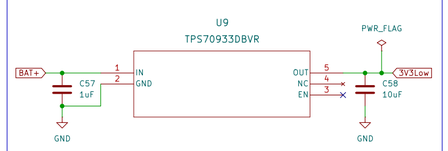
5V Regulator



Info Display, Controls



3.3V Low Power – 150mA MAX
Runs MCU at all times



H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

Sheet: /
File: Pocket–Lab–Bench–Power.kicad_sch

Title: Portable Power Bank/Lab Bench Supply

Size: A2 | Date: 2026–06–30 | Rev: REV1
KiCad E.D.A. 10.0.1 | Id: 1/1